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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester M. Tech (CCT) (CBCS) Degree Examinations

December 2024- January 2025

22CCT131 – MECHANIZATION IN CONSTRUCTION

Duration: 3 Hours

Max. Marks: 100

Note: Answer Two full questions from Unit I and Unit II, One full question from Unit III

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain briefly the difference between standard and special equipment in construction? Also list the advantages and Disadvantages of standard and special equipment.	10	L*2	1	1
	b) Explain in detail advantages and limitations of mechanization in construction industry.	10	L3	1	1, 2
2.	a) Compare the advantages and disadvantages of using excavators versus backhoes.	10	L3	2	1, 2
	b) Explain briefly the ongoing ownership costs of construction equipment? Explain how the ongoing ownership costs of construction equipment can be reduced?	10	L3	2	1, 2
3.	a) Analyze the key parameters which governs the efficiency and productivity of excavators during the earth moving process?	10	L4	2	1, 2
	b) Analyze the factors that influence the selection of a dragline or clamshell for a specific project.	10	L4	2	1, 2
Unit – II					
4.	a) Identify the challenges and opportunities for further mechanization in the aggregate manufacturing industry.	10	L4	3	1, 2
	b) Analyze the advantages and disadvantages of using precast concrete versus site-cast concrete in terms of mechanization.	10	L4	3	1, 2, 7
5.	a) Compare the efficiency and cost-effectiveness of segmental construction with traditional methods.	10	L4	3	1, 2, 11
	b) Calculate the cost savings associated with the mechanization in rebar fabrication?	10	L3	3	1, 2, 7
6.	a) Explain briefly the box pushing technology for tunnelling.	10	L2	4	1, 2, 5
	b) How can trench less technology can be used to install underground cables and utilities? Explain briefly with advantages.	10	L3	4	1, 2, 5
Unit – III					
7.	a) What are the factors to consider when choosing a type of form work? Explain briefly the challenges and opportunities for using formwork in different applications?	10	L3	5	1, 2
	b) Briefly explain the fundamental principles of formwork and scaffolding design?	10	L3	5	1, 2
8.	a) Explain briefly the advantages and disadvantages of precast and prefabricated constructions?	10	L2	5	1, 2, 7
	b) Explain briefly the safety and environmental issues of the construction equipments like Cranes, Bulldozers and Excavators.	10	L2	5	1, 2, 7

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
